

Sai Ravi Teja Gangavarapu

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EDUCATION

Carnegie Mellon University

Master of Software Engineering – Scalable Systems (Robotics, ML Systems, Computer Vision)

Pittsburgh, PA

Aug 2025 – Dec 2026

Mahindra École Centrale

Bachelor of Technology in Computer Science

Hyderabad, India

Aug 2020 – Dec 2023

SKILLS

Languages: C++, Python, CUDA, Go, TypeScript, SQL, x86 Assembly

Robotics: ROS2 / Nav2, motion planning & control (MPPI, MPC, Hybrid A*), SLAM, state estimation & localization, LiDAR / camera / IMU sensor fusion, sim-to-real transfer

Perception & ML: PyTorch, segmentation, detection, diffusion models, TensorRT, ONNX, INT8 quantization, Jetson deployment

Simulation: NVIDIA Isaac Sim, Unity, Gazebo, hardware-in-the-loop, procedural scenario generation, physics simulation

Systems & Infra: Linux, Docker, Kubernetes, CI/CD (GitHub Actions, ArgoCD), AWS, OpenMP, SIMD, performance profiling

Coursework: Planning & Decision-making in Robotics (16-782), ML Systems (15-642), Visual Learning & Recognition (16-824), Diffusion & Flow Matching (10-799), F1TENTH Autonomous Racing

EXPERIENCE

Autonomy Subsystem Lead – CMU Lunabotics (NASA ‘First Steps’ Award)

Pittsburgh, PA

Advised by Prof. Wenshan Wang, CMU Robotics Institute

Jan 2026 – Present

- NASA competition to build a lunar robot that autonomously digs and dumps regolith while evading rocks and craters; as the first first-year team to score core autonomy points, we won an Autonomy Award (NASA’s ‘First Steps’). Implemented the rover’s ROS2 navigation stack (Hybrid A* planning, GPU-optimized MPPI control evaluating thousands of candidate trajectories in ~0.5 ms vs. ~50 ms on CPU, ~100× faster).
- Trained and deployed an instance segmentation model (PyTorch/YOLOv8) for onboard obstacle detection on synthetic Unity data, quantizing FP32 to INT8 via **TensorRT** for the Jetson Orin, cutting model size 65.6% with only ~3% mIoU drop; built a LiDAR + camera fusion pipeline (RANSAC ground-plane removal for obstacles, point-to-pixel projection to localize dig targets) feeding the costmap.
- Architected the autonomous excavation and dumping sequence using sensor feedback and a state machine, validated through 100+ hours of integrated field testing with the controls and avionics teams.

Researcher – AirLab, CMU Robotics Institute (Prof. Sebastian Scherer)

Pittsburgh, PA

Adversarial Robustness & Verification of Robotic Perception

Aug 2026 – Present

- Building an agentic, automated test bench in **NVIDIA Isaac Sim** that auto-generates scenarios and surfaces failure modes in learning-based UAV perception, SLAM, and obstacle-avoidance across nominal, off-nominal, and adversarial conditions.

Autonomy ML Systems Intern – Motional (Level-4 self-driving)

Pittsburgh, PA

Autonomy Subsystems Evaluation

May 2026 – Aug 2026

- Built multimodal evaluation metrics for multiagent trajectory prediction that account for vehicle kinematics, planner behavior, and danger level, bucketing scenarios by maneuver type; helped tune the prediction model for pedestrian-safety scenarios; engineered the data platform (SQL, AWS Athena, Polars) processing ~1 PB of driving logs.

Machine Learning Engineer – CMU MSE Capstone, Siemens Energy

Pittsburgh, PA

Robotic Blade Inspection & Defect Detection

Aug 2026 – Present

- Building an ML system that flags turbine-blade cracks from phased-array ultrasonic scans, and computer vision for a robotic inspection arm; went on-site with technicians, built classification and clustering models for a small, ~10% anomaly-rate dataset.

Earlier: Founding SWE, **Tapsta** (full-stack app, 1,500+ users, LangGraph search, CI/CD, led 3 interns) | SDE Intern, **Hashira.io** (LLM fine-tuning, \$150M+ trading volume) | Research Assistant, Fan Lab (UF) & Mahindra (IEEE CEC 2024)

PROJECTS

CMR Driverless (Carnegie Mellon Racing) | *ROS2, LiDAR-Camera Fusion* – Contributing to the LiDAR-camera cone detection and classification pipeline for an autonomous Formula SAE car.

F1TENTH Autonomous Racing | *ROS2, Motion Planning, Simulation* – Processed raw LiDAR scans for reactive obstacle avoidance (Follow the Gap) and wall following, with path tracking on 1/10-scale vehicle dynamics; completed CMU’s F1TENTH course.

GPU Ray Tracer & Real-Time Renderer | *CUDA, C++, OpenGL* – Built a GPU-accelerated ray tracer in CUDA (1,600x over CPU) with a real-time GPU boids simulation and a modular 3D renderer in C++/OpenGL with custom shaders; wrote OpenMP-parallel CPU algorithms (10x) and hand-tuned x86/SIMD kernels.

ACHIEVEMENTS

Hackathons (6x winner): 3rd, CMU AWAP 2026 (~80 teams) – multi-agent bot with Hungarian assignment and Johnson’s + BFS routing; 1st, NR Hackathon – Raft consensus from scratch; 2nd, TartanHacks 2026 (\$3K)

Leadership & Publication: President, CS Club (2021-23), partnerships with Ubisoft and NVIDIA; IEEE CEC 2024 publication on emotion-aligned music composition; Best Poster Award, Diffusion & Flow Matching course (sketch-to-face synthesis)